VERSALITY PROPAGATES FROM A POINT TO ITS GENERALIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3075-3088 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3090-3153, the complete author proof (62 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_artin.json (pb=2586, stmt 3075-3088, proof 3090-3153).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 045G: sources/stacks-artin/pinned-original-source.tex, source lines 3153–3153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CXM: sources/stacks-artin/pinned-original-source.tex, source lines 2013–2022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0G2J: sources/stacks-artin/pinned-original-source.tex, source lines 2951–2972. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-artin/pinned-original-source.tex, source lines 3153–3153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. Assume*

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ *is representable by algebraic spaces,*
(2) $\mathcal{X}$ *has (RS*),*
(3) $\mathcal{X}$ *is limit preserving.*

Let x be an object of $\mathcal{X}$ over a scheme U of finite type over S. Let $u \rightsquigarrow u_0$ be a specialization of finite type points of U such that x is versal at u_0. Then x is versal at u.

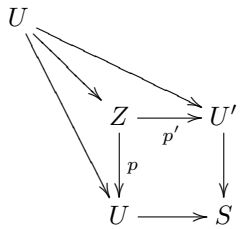
Proof. After shrinking U we may assume U is affine and U maps into an affine open $\mathrm{Spec}(\Lambda)$ of S. If x is not versal at u then we may pick $x \to y$ lying over $U \to T$ as in Lemma 0G2J. Write $U = \mathrm{Spec}(R_0)$ and $T = \mathrm{Spec}(R)$. The morphism $U \to T$ corresponds to a surjective ring map $R \to R_0$ whose kernel is an ideal of square zero. By assumption (3) we get that y comes from an object x' over $U' = \mathrm{Spec}(R')$ for some finite type Λ-subalgebra $R' \subset R$. After increasing R' we may and do assume that $R' \to R_0$ is surjective, so that $U \subset U'$ is a first order thickening. Thus we now have

$$x \to y \to x' \text{ lying over } U \to T \to U'$$

By assumption (1) there is an algebraic space Z over S representing

$$(Sch/U)_{fppf} \times_{x,\mathcal{X},x'} (Sch/U')_{fppf}$$

see Algebraic Stacks, Lemma 045G. By construction of 2-fibre products, a V-valued point of Z corresponds to a triple (a, a', α) consisting of morphisms $a : V \to U$, $a' : V \to U'$ and a morphism $\alpha : a^*x \to (a')^*x'$. We obtain a commutative diagram



The morphism $i : U \to Z$ comes the isomorphism $x \to x'|_U$. Let $z_0 = i(u_0) \in Z$. By Lemma 0CXM we see that $Z \to U'$ is smooth at z_0. After replacing U by an affine open neighbourhood of u_0, replacing U' by the corresponding open, and replacing Z by the intersection of the inverse images of these opens by p and p', we reach the situation where $Z \to U'$ is smooth along $i(U)$. Since $u \rightsquigarrow u_0$ the point u is in this open. Condition (3) of Lemma 0G2J is clearly preserved by shrinking U (all of the schemes U, T, U' have the same underlying topological space). Since $U \to U'$ is a first order thickening of affine schemes, we can choose a morphism $i' : U' \to Z$ such that $p' \circ i' = \mathrm{id}_{U'}$ and whose restriction to U is i (More on Morphisms of Spaces, Lemma 04AM). Pulling back the universal morphism $p^*x \to (p')^*x'$ by i' we obtain a morphism

$$x' \to x$$

lying over $p \circ i' : U' \to U$ such that the composition

$$x \to x' \to x$$

is the identity. Recall that we have $y \to x'$ lying over the morphism $T \to U'$. Composing we get a morphism $y \to x$ whose existence contradicts condition (3) of Lemma 0G2J. This contradiction finishes the proof. □